Ans. (c) : The exotic cold water fishes introduced in India include trouts, mirror carps, crucian carps and tenches (Tinca). The fist attempt to transplant trout in India was made at Nilgiris in 1907. In the Himalayas, trout has been transplanted in Kashmir and Himachal Pradesh, in Garhwal Himalayas, Arunachal Pradesh, Nagaland, Meghalaya and in certain areas of Nepal.
Tinca also called the doctorfish. It was first introduced into the Ooty lake as early as 1874.

39. Which one of the following soil types of India is rendered infertile by the presence of excess iron ?

(a) Desert sand (b) Alluvial
(c) Podzolic (d) Lateritic

Ans. (d) : Lateritic soil is rendered infertile by the presence of excess iron. Lateritic soil are rich in iron and aluminum, low in silica, chemically acidic, and exhibit a soil profile different from that of other soils. Lateritic soils are one of important soils and are widespread in tropical areas and subtropical climates.

40. The ideal climatic conditions for the cultivation of rice are

(a) rainfall above 100 cm, temperature above 25°C
(b) cool and moist climate for the entire crop period
(c) rainfall below 100 cm, temperature below 25°C
(d) warm and dry climate during the entire crop period

Ans. (a) : The ideal climatic conditions for the cultivation of rice are rainfall above 100 cm, temperature above 25^0C. Rice crop needs a hot and humid climate. It is best suited to regions which have high humidity, prolonged sunshine and an assured supply of water.
In India rice is grown under widely varying conditions of altitude and climate. Rice cultivation in India extends from 8 to 35^0N latitude and from sea level to as high as 3000 meters.

41. The density of cattle population per 100 hectares of gross cropped area is the highest in

(a) Bihar (b) Haryana
(c) Madhya Pradesh (d) Uttar Pradesh

Ans. (b) : The density of cattle population per 100 hectares of gross cropped area is the highest in Haryana.
Cattle population in Haryana is 11.85%.

42. Even though the Railways are the most pervasive mode of transport in India, it is the road transport which received the maximum impetus in most of the post-independence period.
Which of the following are the reasons for this ?
I. Railways are cheap to operate but the associated capital costs are very high.
II. Given the geographic spread of the human settlements, it is unrealistic that the railways alone can serve the needs of transport.
III. The per unit (tonne/km) cost of transport is cheaper in road compared to rail transport.
IV. Given the indivisible nature of the Railways, it is not always convenient for the population to avail it with the same ease as private cars, buses or two wheelers.
Select the correct answer from the codes given below
Codes:
(a) I, II and III (b) I, II and IV
(c) I, III and IV (d) II, III and IV

Ans. (b) : Even though the Railways are the most pervasive mode of transport in India. It is the road transport which received the maximum impetus in most of the post independence period because of Railways are cheap to operate but the associated capital costs are very high & the geographic spread of the human settlements. It is unrealistic that the railways alone can serve the needs of transport.
The indivisible nature of the Railways. It is not always convenient for the population to avail of it with the same ease as private cars, buses or two-wheelers.

43. Which one of the following is the most important reason for the establishment of the Vikram Sarabhai Space Centre at Thumba?

(a) The sea is near to the place in the East, West and South directions
(b) It is a place near to a city in India, which is nearest to the Geographic Equator
(c) It is situated on the Geomagnetic Equator
(d) Outer space near Thumba exhibits queer atmospheric phenomena

Ans. (c) : Vikram Sarabhai Space Centre (VSSC), Thiruvananthpuram, is the lead centre of ISRO responsible for the design and development of launch vehicle technology.
The major programmes at VSSC include Polar Satellite Launch Vehicle (PSLV), Geosynchronous Satellite Launch Vehicle (GSLV) and Rohini Sounding Rockets as well as the development of GSLV Mk-III. The most important reason to establishment of the VSSC at Thumba, is that it is situated on the Geomagnetic Equator.

44. In recent years, there has been some concern over the threat posed by the Mathura Oil Refinery and the thermal power plants to the Taj Mahal in Agra. The scientific basis of any possible damage to the Taj is mainly

(a) stratospheric ozone and the chlorofluorocarbons which destroy it
(b) acid precipitation and tropospheric ozone
(c) increasing levels of atmospheric CO_2 which produce the greenhouse effect
(d) ultraviolet radiation (240260 nanometers wavelength) and the fumes from the heavily polluted Yamuna river

Ans. (b) : The Taj Mahal, a mausoleum constructed in white Marble by the mughal emperor Shahjahan in the beloved memory of his wife, has been accredited as being one of the seven wonders of the world, by millions of people worldwide.
The most damaging of all forms of pollution on the marble of Taj have been air and water pollution. Effluents are released from over 1700 factories that exist in and around Agra, in the form of gases such as CO, CO_2, SO_2, chlorofluoro carbons, nitrogen oxides particulate matter etc. and in recent years, there has been some concern over the threat posed by the Mathura Oil Refinary and the thermal power plants to the Taj Mahal in Agra. The scientific basis of any possible damage to the Taj is mainly by acid precipitation and tropospheric ozone.

WORLD GEOGRAPHY

45. The time taken by the sun to revolve around the centre of our galaxy is

(a) 25 million years (b) 100 million years
(c) 250 million years (d) 500 million years

Ans. (c) : Our Milky Way galaxy is a big place. Even at this blazing speed, it takes the sun approximately 225-250 million years to complete one journey around the galaxy's center. This amount of time-the time it takes us to orbit the center of the galaxy is sometimes called a cosmic year.

46. **The length of its day and the tilt of its axis are almost identical to those of the earth'. This is true of**

(a) Uranus (b) Neptune
(c) Saturn (d) Mars

Ans. (d) : Mars has two poles like Earth, it is inclined at an angle of 20 degrees to its axis, due to which the seasons change here like Earth.

47. Consider the map given below :



The numbers marked 1, 2, 3 and 4 respectively indicate the islands of

(a) the Bahamas, Azores, Falkland and Cape Verde
(b) the Bahamas, Azores, Cape Verde and Falkland
(c) Azores, Cape Verde, the Bahamas and Falkland
(d) Cape Verde, the Bahamas, Falkland and Azores

Ans. (c) : From the Map, the numbers marked 1, 2, 3 & 4 respectively indicate the islands of Azores, Cape Verde, the Bahamas and Falkland.

48. The greatest diversity of animal and plant species occurs in

(a) temperate deciduous forests
(b) tropical moist forests
(c) heavily polluted rivers
(d) deserts and savannas

Ans. (b) : Tropical and subtropical moist broadleaf forests generally found in large, discontinuous patches centered on the equatorial belt and between the Tropics of cancer and Capricorn.
These trees number in the thousands and contribute to the highest levels of species diversity in any terrestrial major habitat type. In general, biodiversity is highest in the forest canopy.
These forests are home to more species than any other terrestrial ecosystem: Half of the world's species may live in these forests.

49. The acceleration due to gravity of a catastrophic earthquake will be

(a) > 550 cm/sec2
(b) > 750 cm/sec2
(c) > 950 cm/sec2
(d) > 980 cm/sec2